Umberto Domenico Ciccia

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Experience

Systems Development Engineer, **Amazon Web Services (AWS)** – Dublin, Ireland Nov 2025 – Present

- Eliminated 84 false-positive SEV-2 pages over 6 months by tracing same-AZ DNS availability check failures to NAT hairpin behavior in a 3-AZ setup and fixing the monitoring configuration
- Launched Amazon CloudFront for AWS European Sovereign Cloud general availability, building AWS CDK infrastructure as code, deployment pipelines, service APIs, CloudWatch alarms, and on-call operations
- Saved ~4 engineering weeks by building Python AI-agent automation that refactored 184 Route 53 pipelines (AWS CDK, TypeScript, Ruby) in 120 code reviews, increasing fresh-artifact pipelines from 80% to 98%
- Delivered Route 53 DNSSEC in AWS European Sovereign Cloud 3 weeks early, unblocking a key customer migration, by building Java APIs, AWS CDK key rotation, CloudWatch alarms, canaries, and console support
- Delivered CIDR-based DNS routing 3 months ahead of schedule, implementing infrastructure, API, and console changes that let customers route DNS queries by client IP range
- Built a Python Model Context Protocol (MCP) server letting AI agents self-serve customer-metadata redaction in logs and metrics through a partner team distributed service (Lambda, SQS, SNS, Aurora, DynamoDB)
- Resolved on-call incidents for Route 53, AWS WAF, and AWS Shield, writing root-cause analyses and runbooks to prevent recurrence, and validated peer changes with unit, integration, contract, and canary tests
- Led sprint planning and agile delivery as team Scrum Master for 2 quarters

Cloud Software Engineer Consultant, **NTT Data** – Cosenza, Italy Jan 2025 – Oct 2025

- Built an Amazon EKS cluster and replaced manual Bash deployments with Jenkins pipelines automating Terraform, ECR image builds, approval gates, and rollback on CloudWatch error-rate and p99 latency alarms
- Cut estimated infrastructure costs by 30% versus original designs by building a Quarkus AI agent (Vertex AI SDK) that analyzes architectures on the NTT KUMO multi-cloud platform and recommends optimized configurations
- Automated Istio policy management across remote Kubernetes clusters by building a Quarkus microservice on NTT KUMO with Crossplane, replacing manual per-cluster configuration
- Migrated application data from on-premises MariaDB to Amazon Aurora by configuring AWS Database Migration Service (DMS) full load and ongoing change replication
- Built GCP ingestion pipelines processing 10k+ daily records for Coop Italia with Dataflow, Compute Engine, and Cloud Storage, improving data availability and release reliability
- Won an AWS Jam competition representing NTT Data

Software Engineer, **Vision One** – Recanati, Italy July 2024 – Dec 2024

- Cut scheduling time by 80% by developing an organization-wide on-call scheduling system in C#, .NET, and Blazor with rule-based shift assignment, availability validation, and conflict reassignment
- Reduced lines of code by 60% through refactoring and added xUnit tests reaching 80% coverage

Software Engineer, **Caliò Informatica** – Cosenza, Italy Jan 2024 – July 2024

- Cut API costs by 80% and improved performance by 60% in an AI-powered visual workflow builder by eliminating redundant requests and combining calls, validated with before and after profiling

Skills

Cloud and Infrastructure: AWS (CDK, CloudFront, CloudWatch, Route 53, WAF, Shield, Lambda, SQS, SNS, Aurora, DynamoDB, DMS, EKS, ECR), GCP (Vertex AI, Dataflow), Kubernetes, Istio, Crossplane, Terraform, Jenkins, Git, DNS

Languages and Frameworks: Python, Java, TypeScript, C#, Ruby, Bash, Quarkus, .NET, Blazor, xUnit

Education

Università della Calabria – Bachelor of Science in Computer Engineering – GPA: 4.0 2024